Experiment 7:

Write a program to implement Logistic Regression (LR) algorithm in python

Python Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
new_data = [[5.1, 3.5, 1.4, 0.2]]
predicted_class = model.predict(new_data)
print("Actual values:", y_test)
print("Predicted values:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("New input:", new_data[0])
print("Predicted class:", predicted_class[0])
print("Predicted species:", iris.target_names[predicted_class[0]])
```

Output:

```
Actual values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Predicted values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy: 1.0
New input: [5.1, 3.5, 1.4, 0.2]
Predicted class: 0
Predicted species: setosa
```